LARGE SCALE DIGITAL DRIVING OF PROGRAMMABLE PHOTONIC CHIPS

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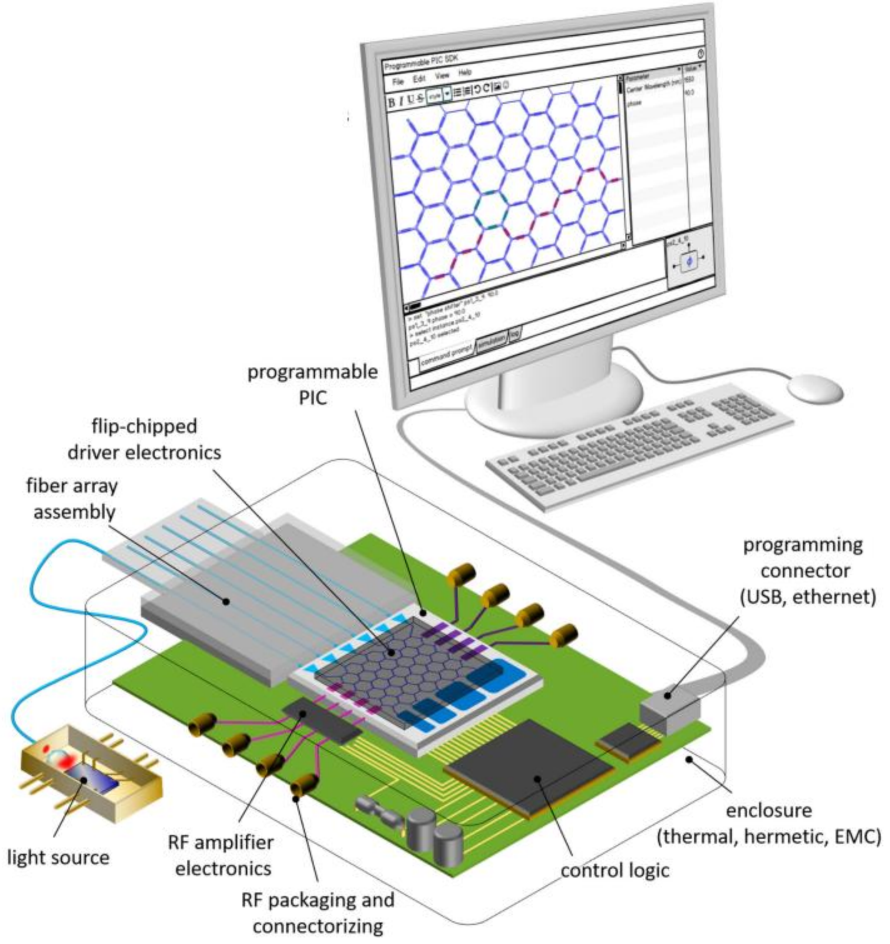
Thesis Story



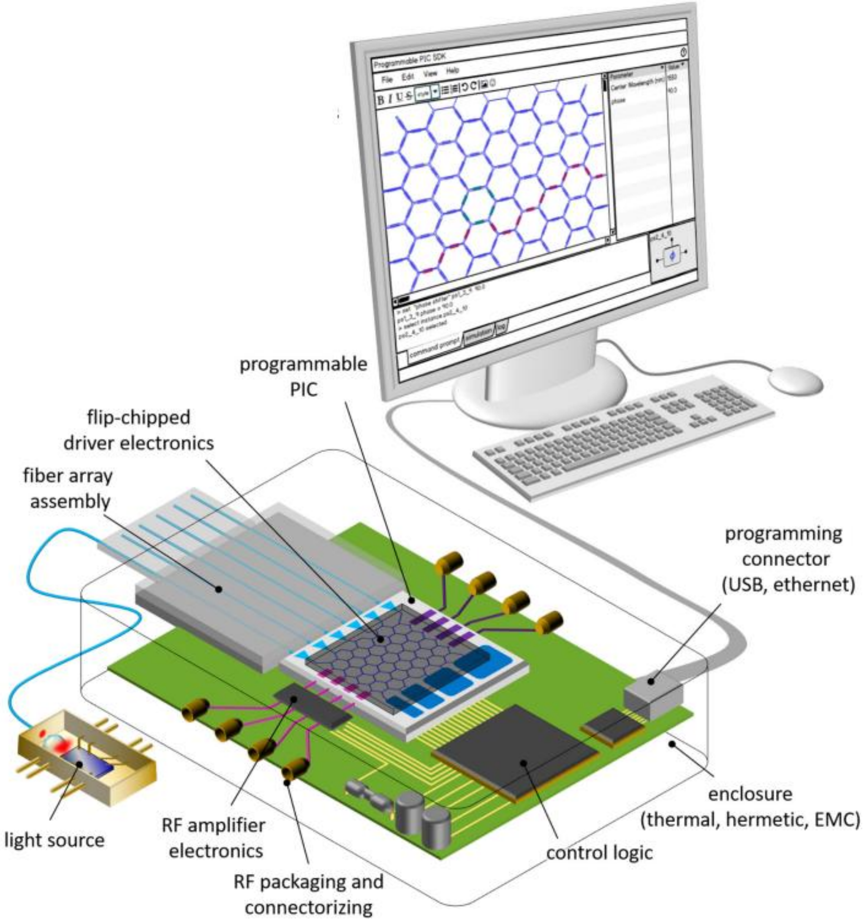
- Photonics are more electronically controlled.

- How to drive and control all the electronics.

  - Digital Driving.
  - Matrix addressing configuration.

Programmable Photonics: An Opportunity for an Accessible Large-Volume PIC Ecosystem

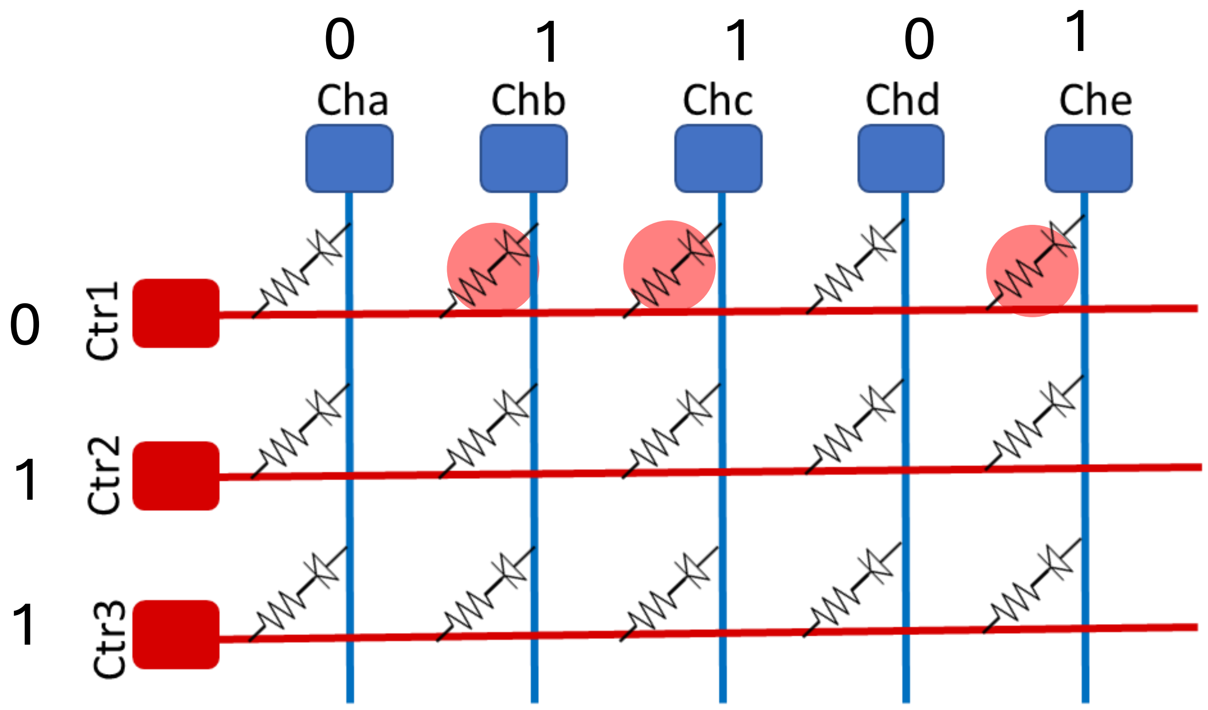
Overview



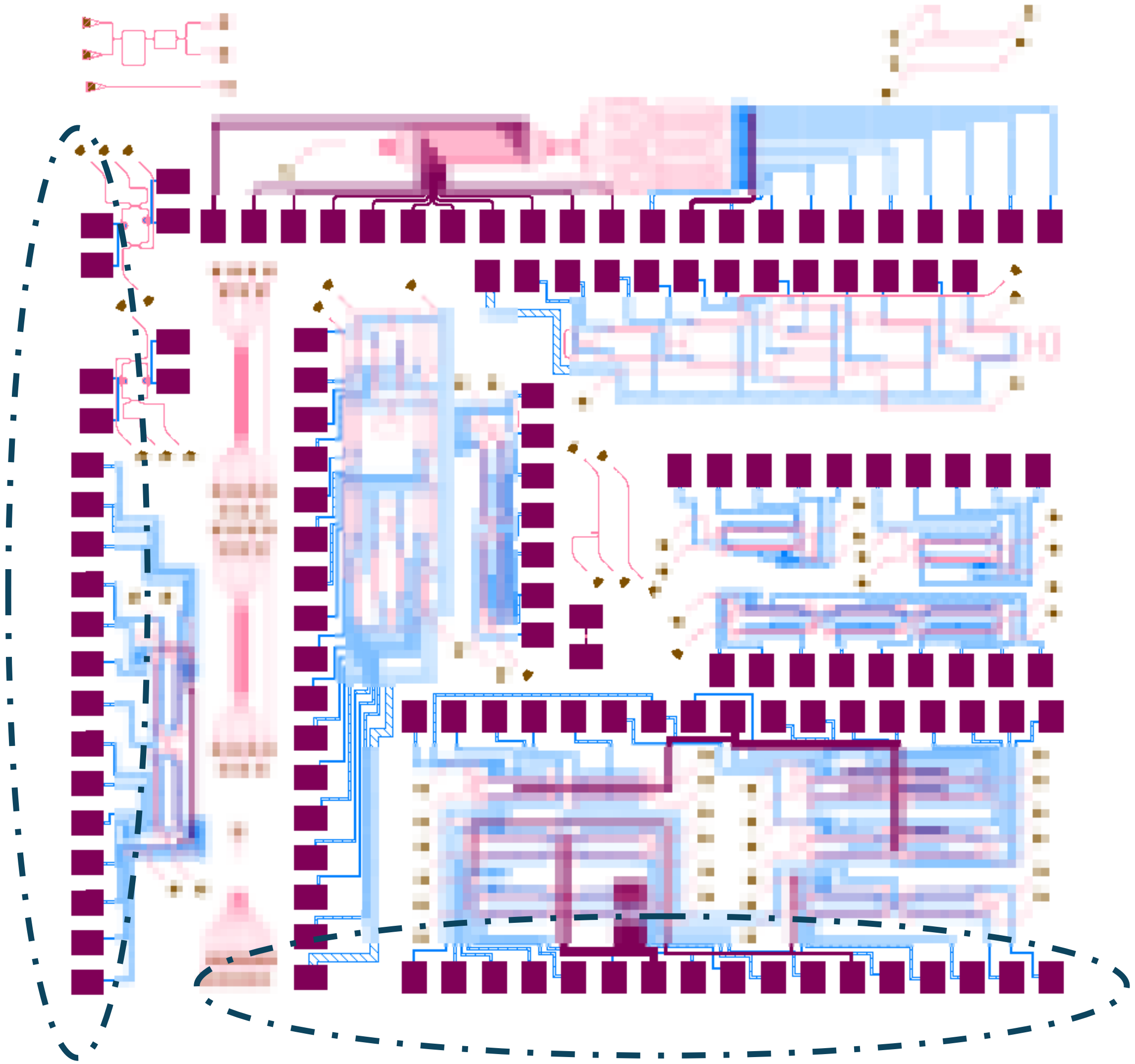
- Programmable photonics circuits.
- Mach Zehnder Interferometry (MZI) is main building unit.
- Light controlled by Optical Phase Shifter heaters.

Programmable Photonics: An Opportunity for an Accessible Large-Volume PIC Ecosystem

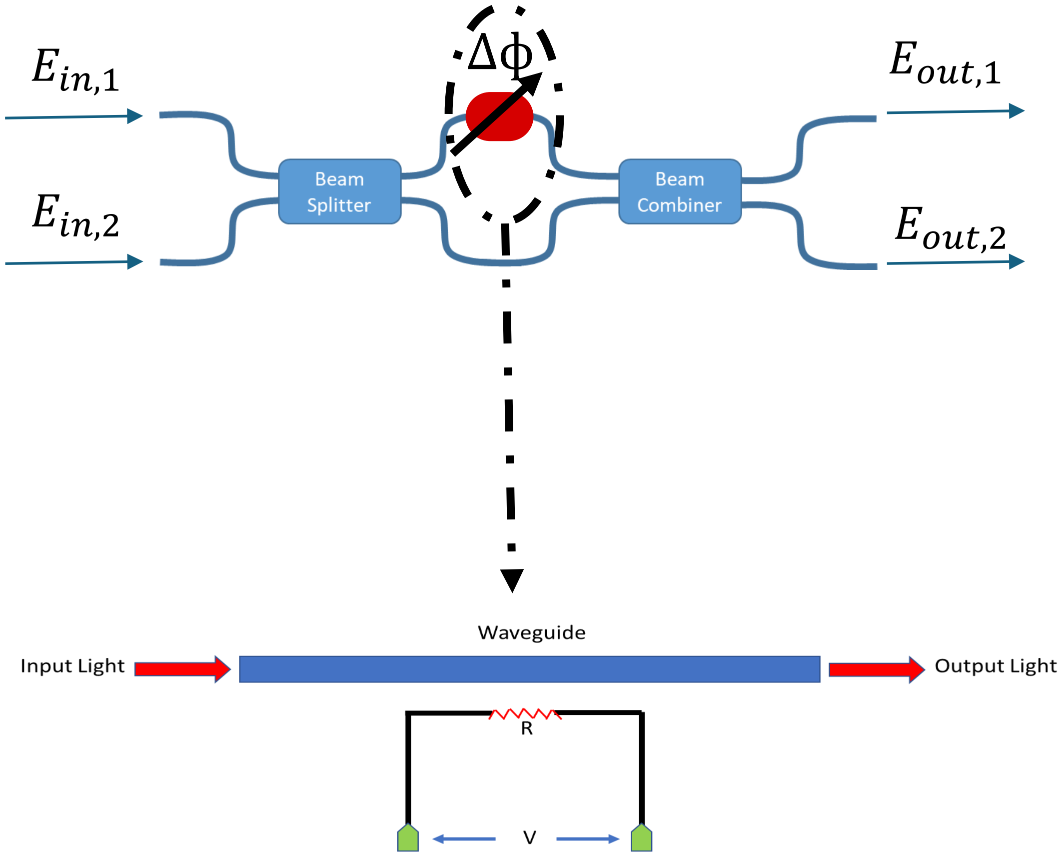
Overview

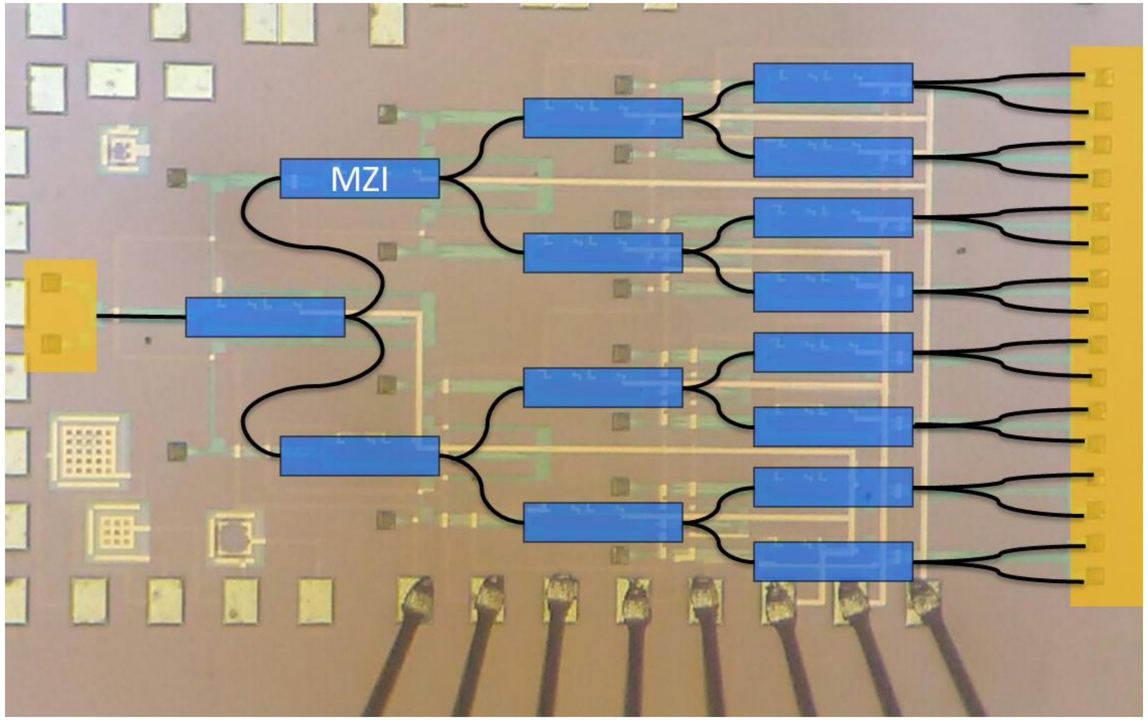


Only N + M pads needed instead of N x M



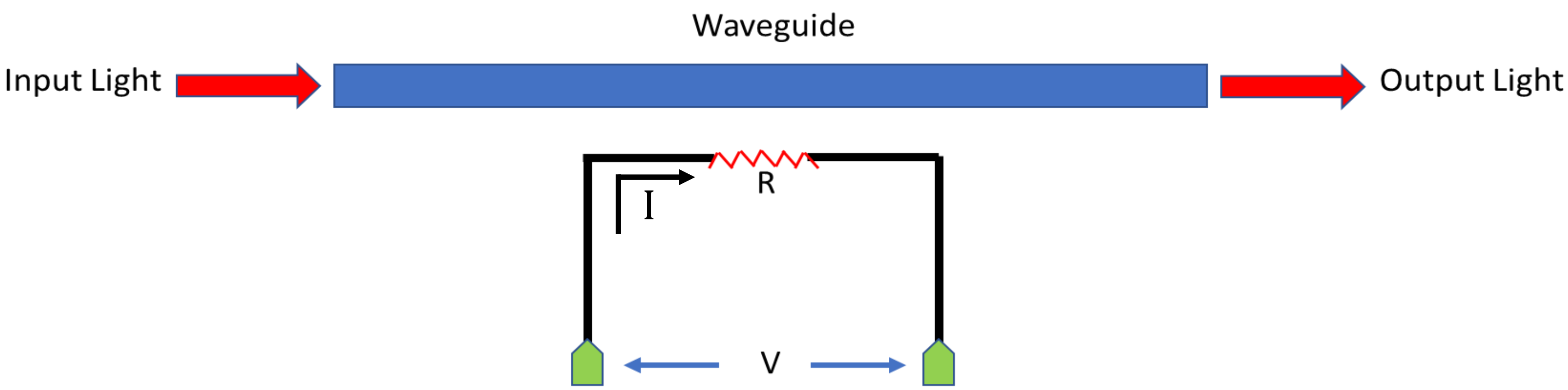
Mach Zehnder Interferometry (*MZI*)





Light controlled by Optical Phase Shifter heaters.

Phase Shifter Heaters

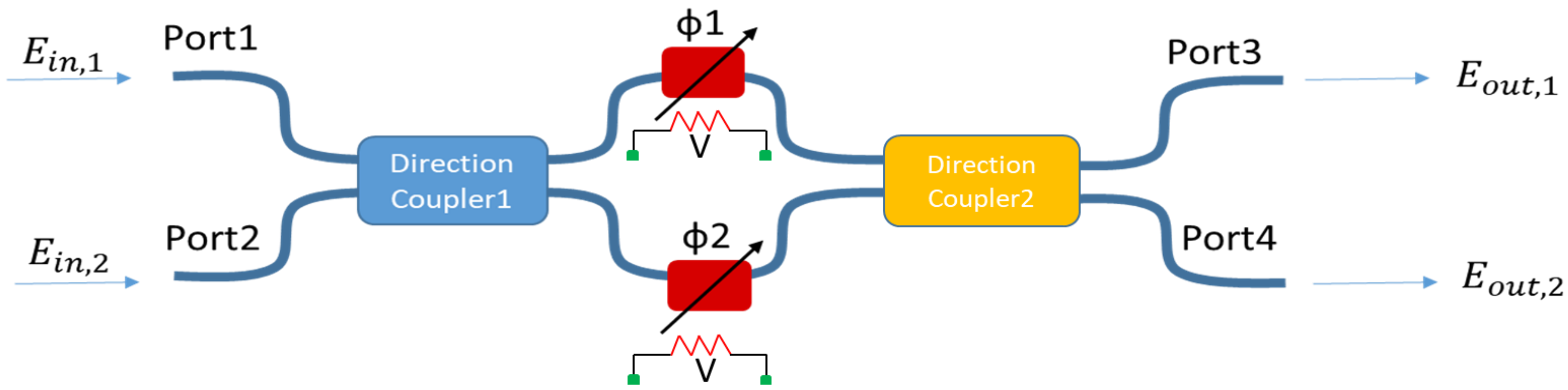


$$P \propto (V^2 / \mathrm{R}) \propto \Delta T \propto \Delta n \propto \Delta\phi$$

$$\text{Silicon thermo-optic coefficient } = dn/dT = 1.8 * 10^{-4}\ K^{-1}$$

$$\Delta\phi = \Delta\beta\ * L = \frac{2\pi\Delta n\ * L}{\lambda}$$

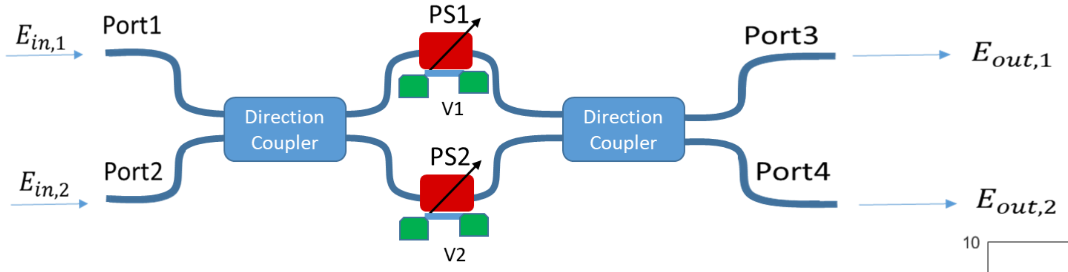
Mach-Zehnder Interferometer Phase Tuning



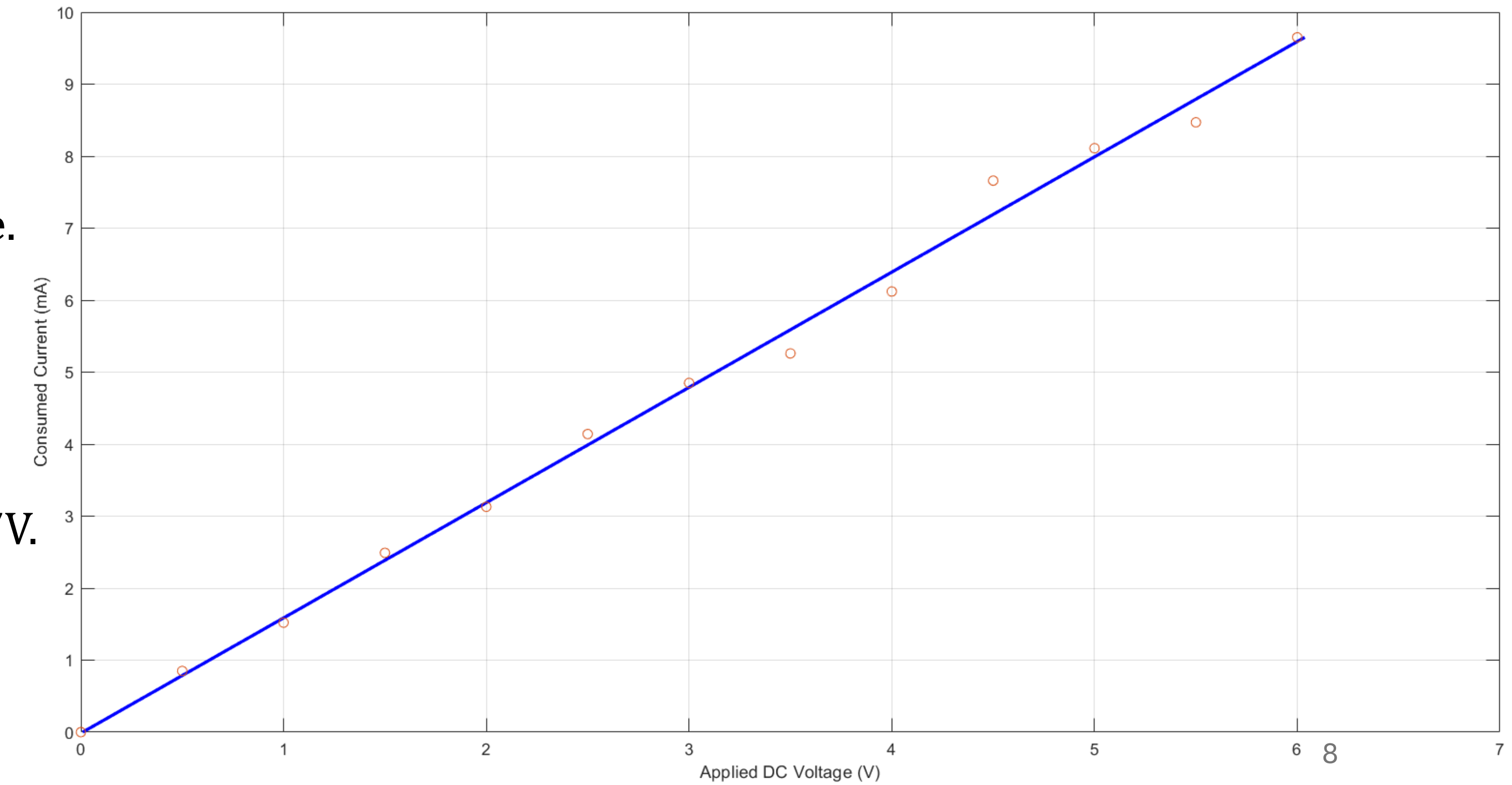
$$\begin{bmatrix} E_{out,1} \\ E_{out,2} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -j\frac{1}{\sqrt{2}} \\ -j\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} e^{-j\phi_1} & 0 \\ 0 & e^{-j\phi_2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -j\frac{1}{\sqrt{2}} \\ -j\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} E_{in,1} \\ E_{in,2} \end{bmatrix}$$

$$\left|\frac{E_{out,1}}{E_{in,1}}\right|^2 = sin^2\left(\frac{\phi_2-\phi_1}{2}\right) \ \& \ \left|\frac{E_{out,2}}{E_{in,1}}\right|^2 = cos^2\left(\frac{\phi_2-\phi_1}{2}\right)$$

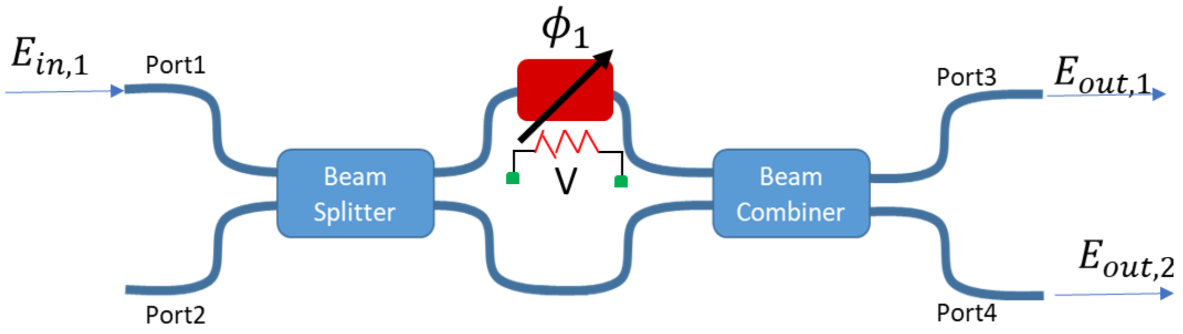
Digital driving for the Optical Phase Shift Heater



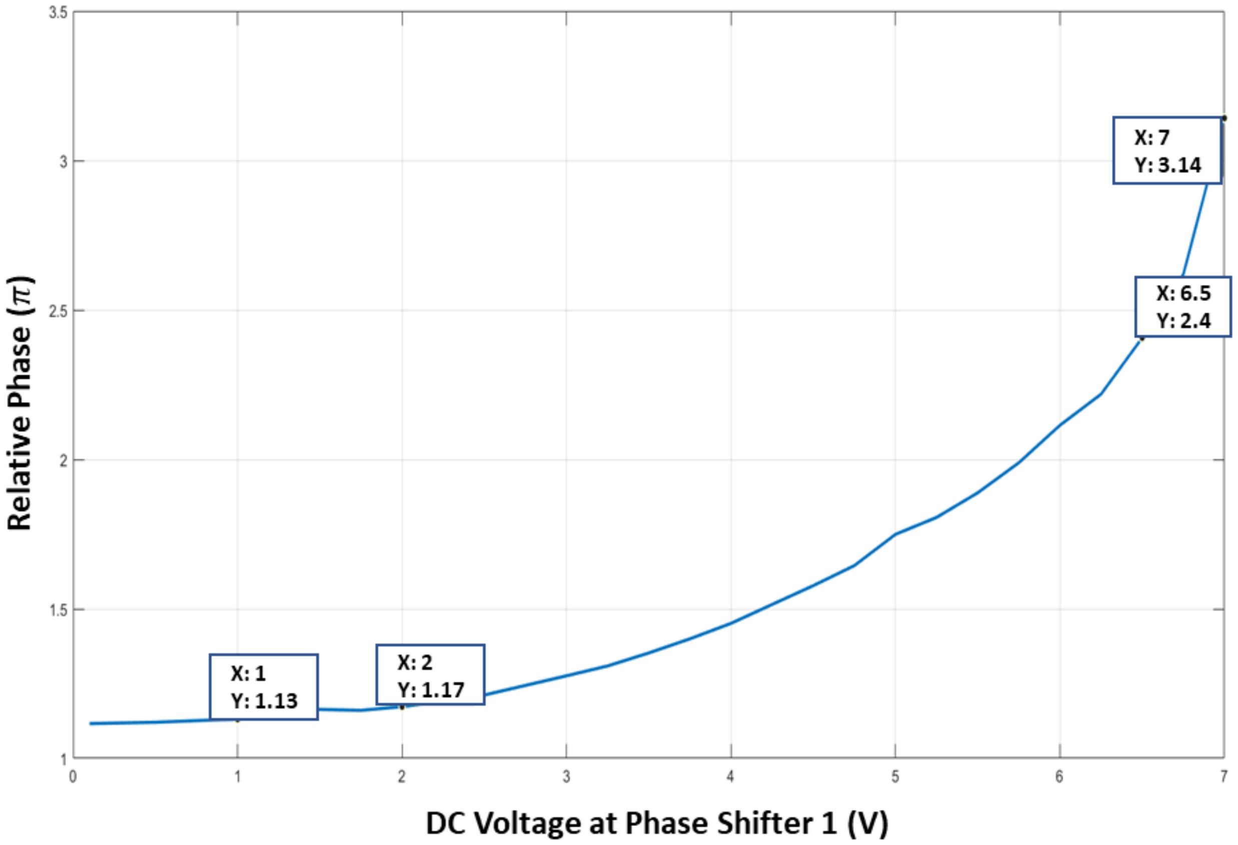
- I-V characterization for Heater resistance.
- R1 = 560 Ω & R2 = 300 Ω
- Max Voltage to apply (Characterized) = 7V.



Mach-Zehnder Interferometer Phase Tuning



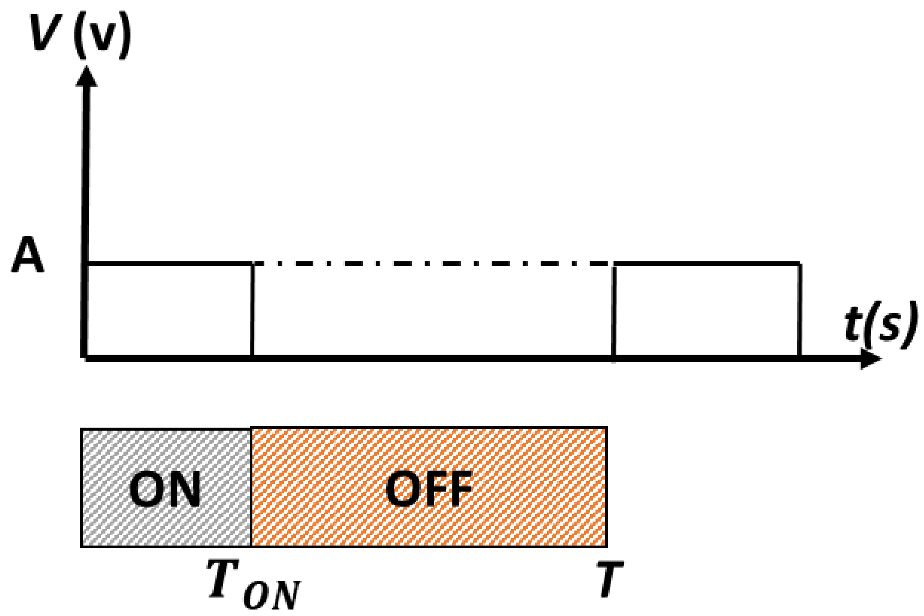
$$P \propto V_{DC}^2 / \mathrm{R} \propto \Delta T \propto \Delta\phi$$



Quadratic response issues for phase differences $\Delta\phi$ at higher voltages!

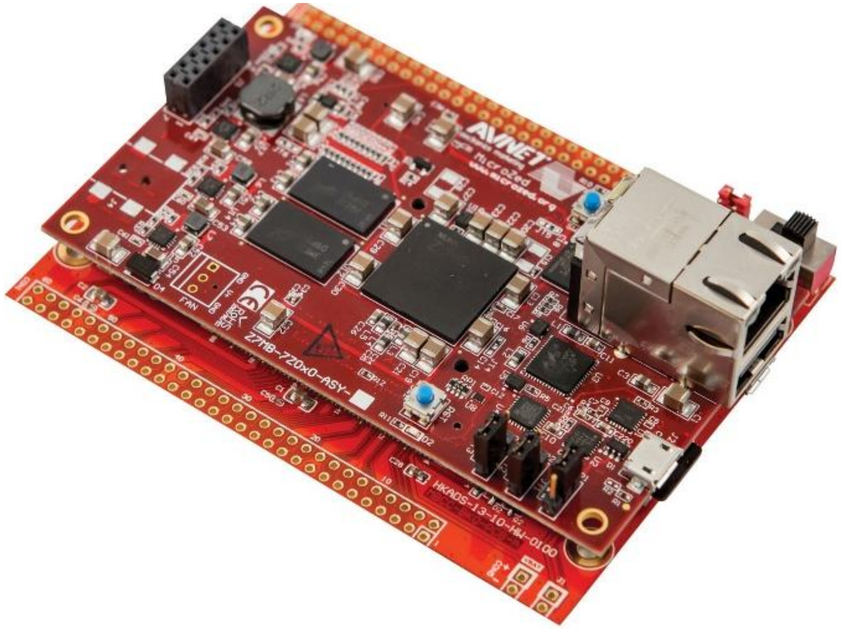
<u>Solution</u>? AC voltage (i.e., *PWM* signal)

Pulse Width Modulation signal (PWM)



$$P_{avg} = \frac{A^2}{R} * Duty\ Cycle\ (\frac{T_{ON}}{T})$$

FPGA to generate PWM signal



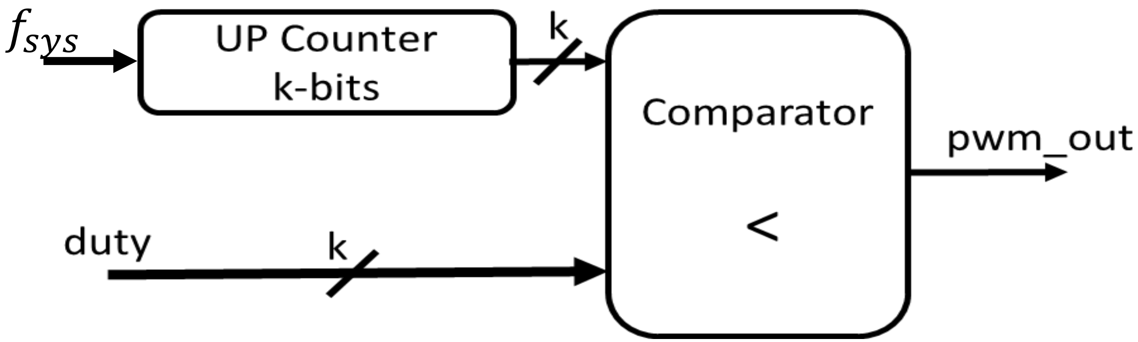
- Field Programmable Gate Array (*FPGA*).
- FPGA used: Xilinx MicroZed (z7010) board.

FPGA Board - Zynq

PWM Design Parameters

- Duty cycle: (0% -> 100 %)

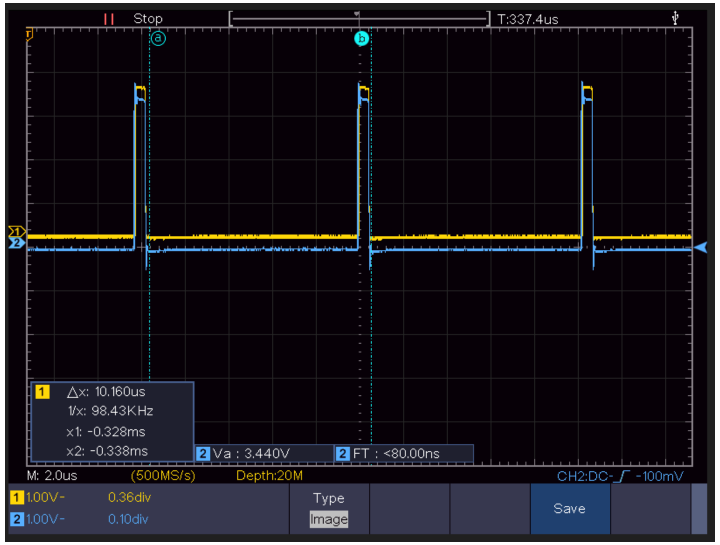
- Resolution: (8-bits: $\frac{0}{256}, \frac{1}{256}, \ldots, \frac{255}{256}, \frac{256}{256}$)



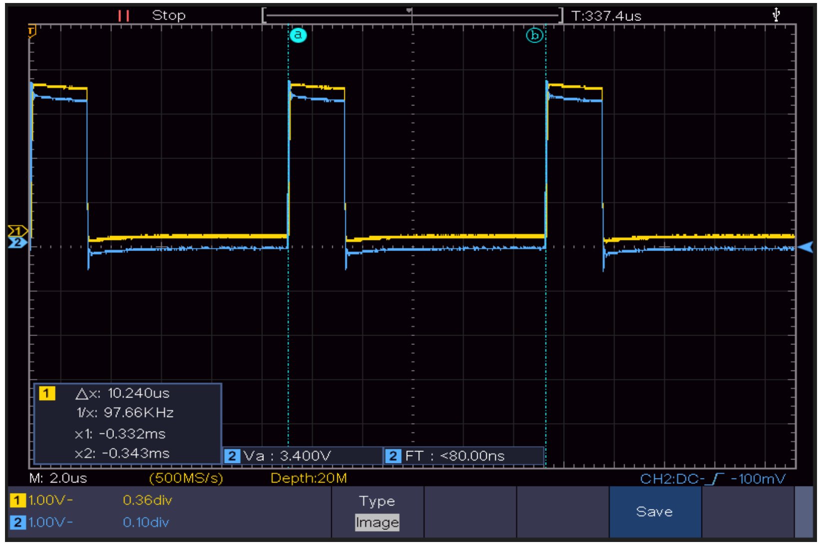
8-bits: $\frac{0}{256}(0), \frac{1}{256}, \frac{2}{256}$,,, , ,, ,, $\frac{255}{256}, \frac{256}{256}$ (=1)

- Switching Frequency $f_{PWM} = \frac{1}{t_{PWM}} = \frac{f_{sys}}{2^k} = \frac{100 * 10^6}{2^8} = 390\ KHz.$

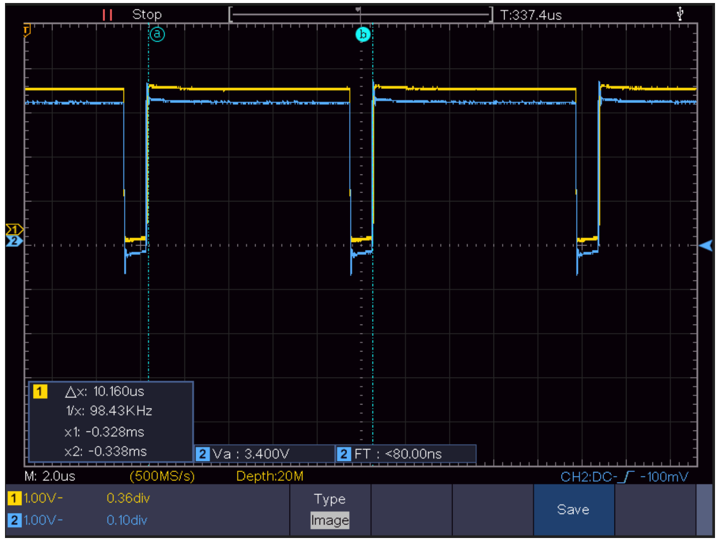
FPGA PWM Output signal (10-bits resolution)



Outputs: 3.3V
Duty cycle=10%
Freq.= 98KHz



Outputs: 3.3V
Duty cycle=25%
Freq.= 98KHz

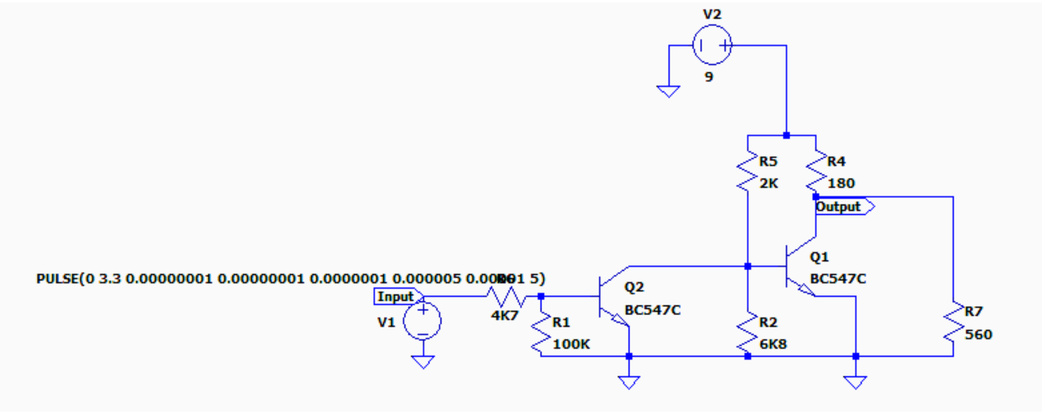


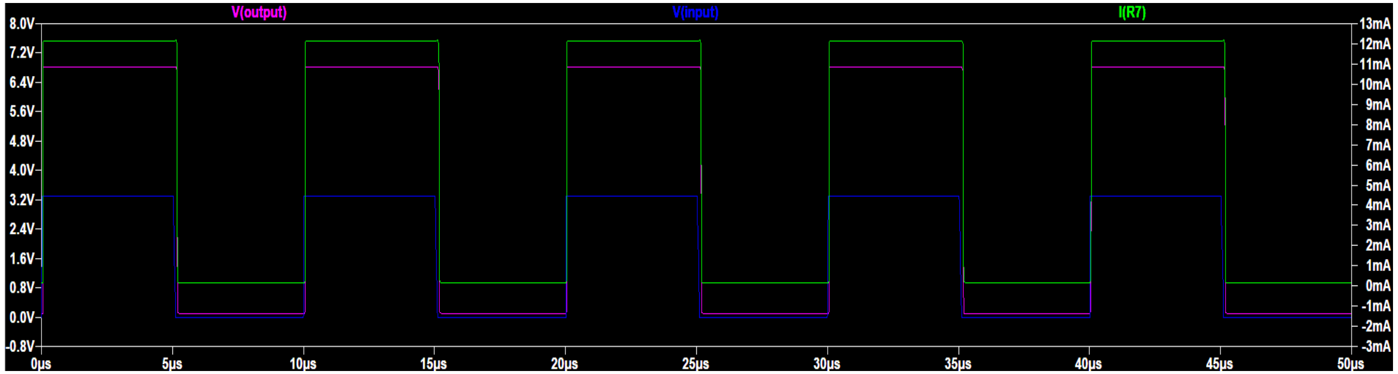
Outputs: 3.3V
Duty cycle=90%
Freq.= 98KHz

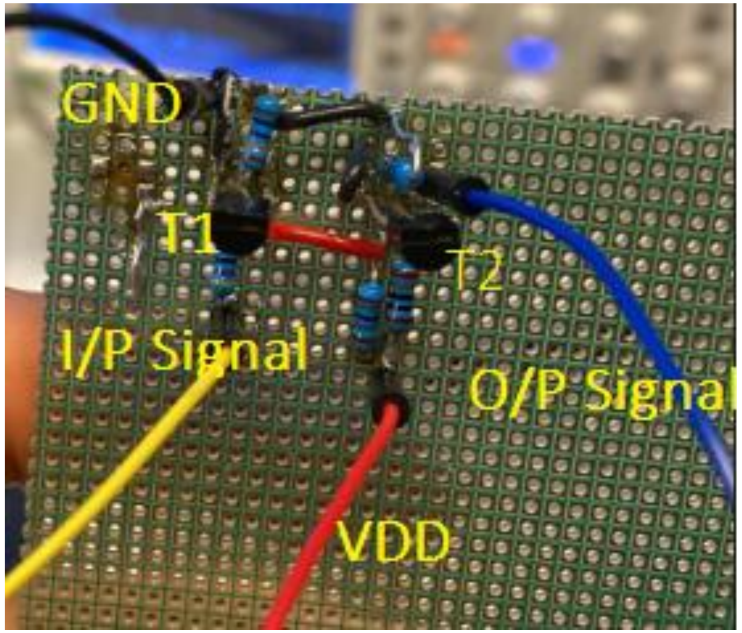
FPGA Outputs Voltage: 3.3V with no enough current for phase shifter heater.

<u>Solution</u>? *Designing Booster Circuit*

Boosting FPGA Outputs

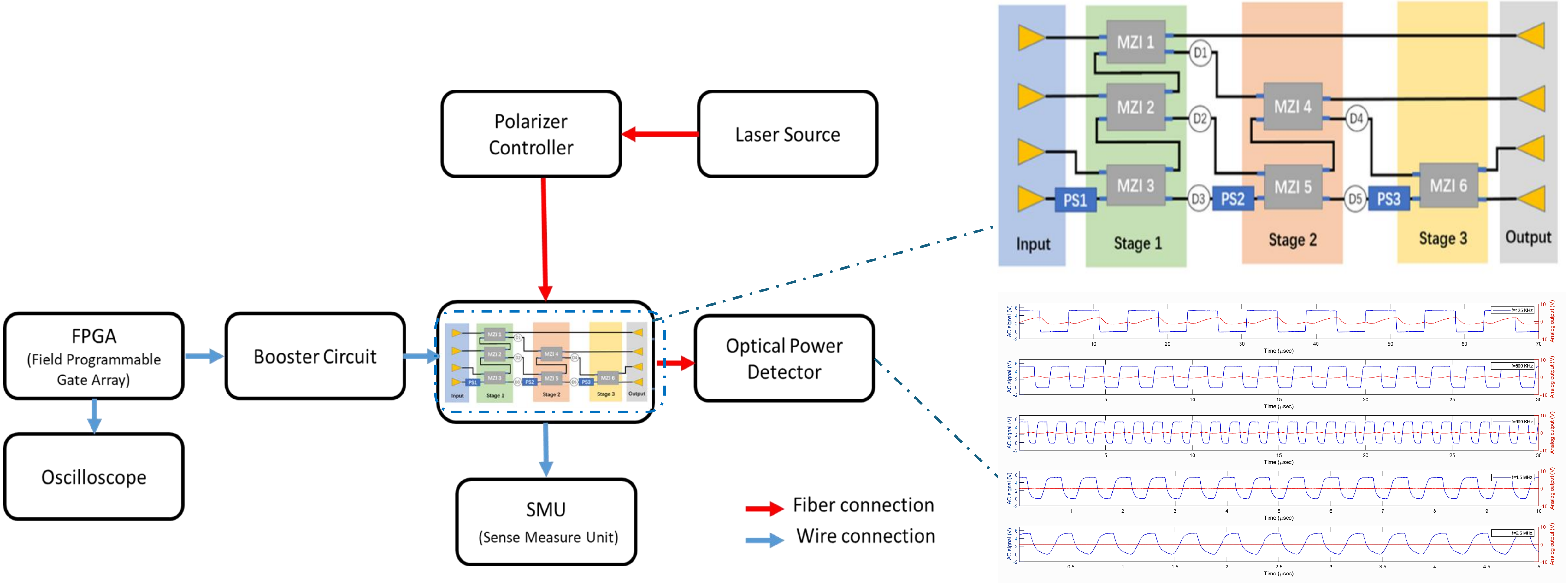






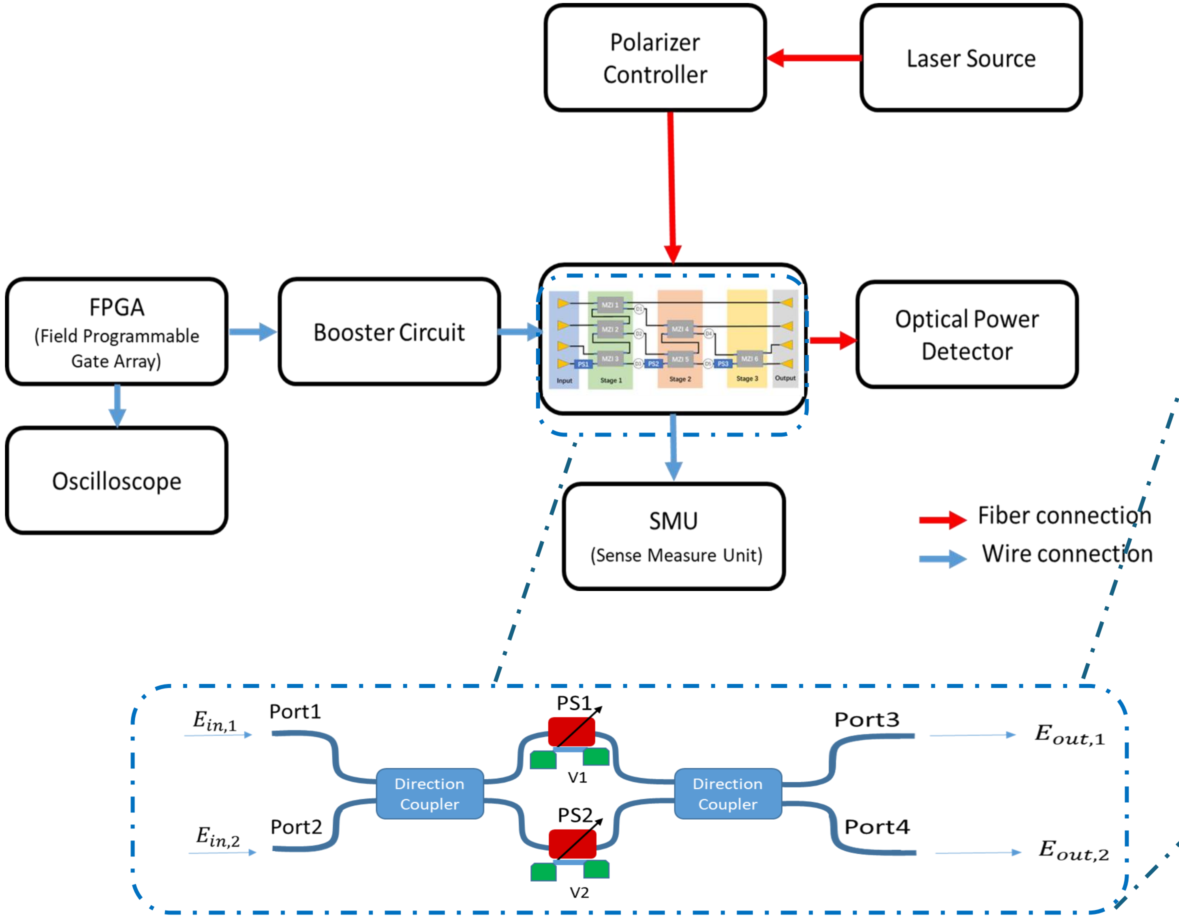
Maximum Switching Limitations

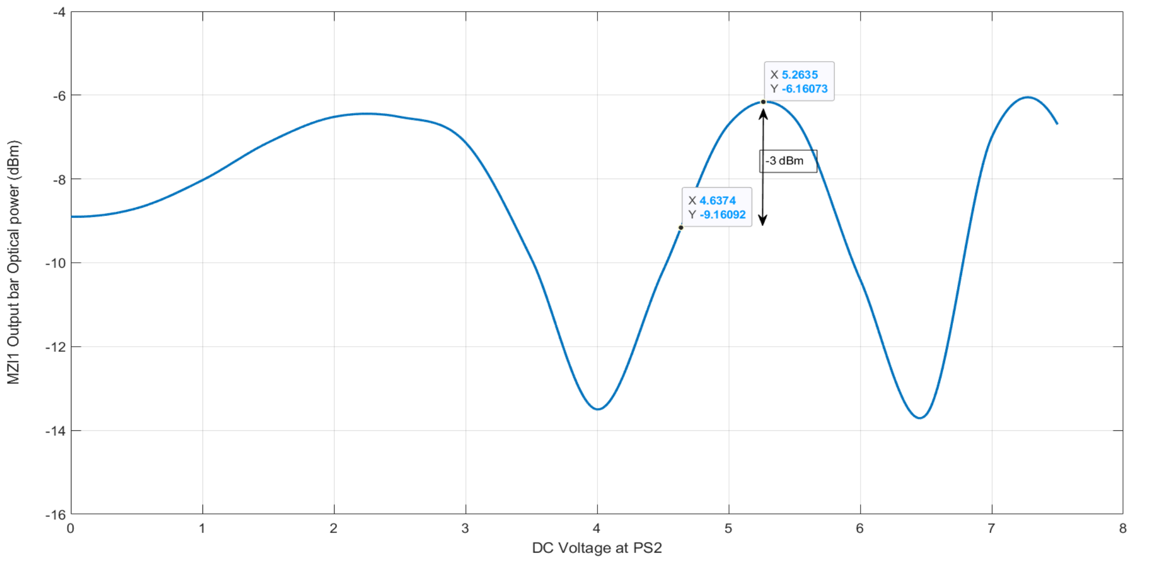
Setup

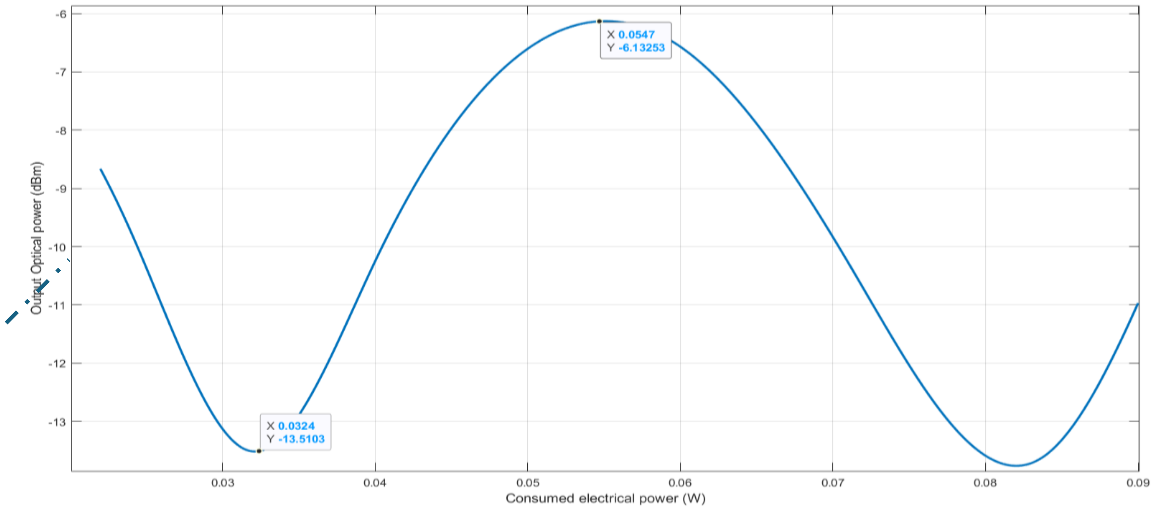


Polarizer Controller
Laser Source
FPGA (Field Programmable Gate Array)
Booster Circuit
Optical Power Detector
Oscilloscope
SMU (Sense Measure Unit)
Fiber connection
Wire connection
MZI 1
MZI 2
MZI 3
MZI 4
MZI 5
MZI 6
D1
D2
D3
D4
D5
PS1
PS2
PS3
Input
Stage 1
Stage 2
Stage 3
Output

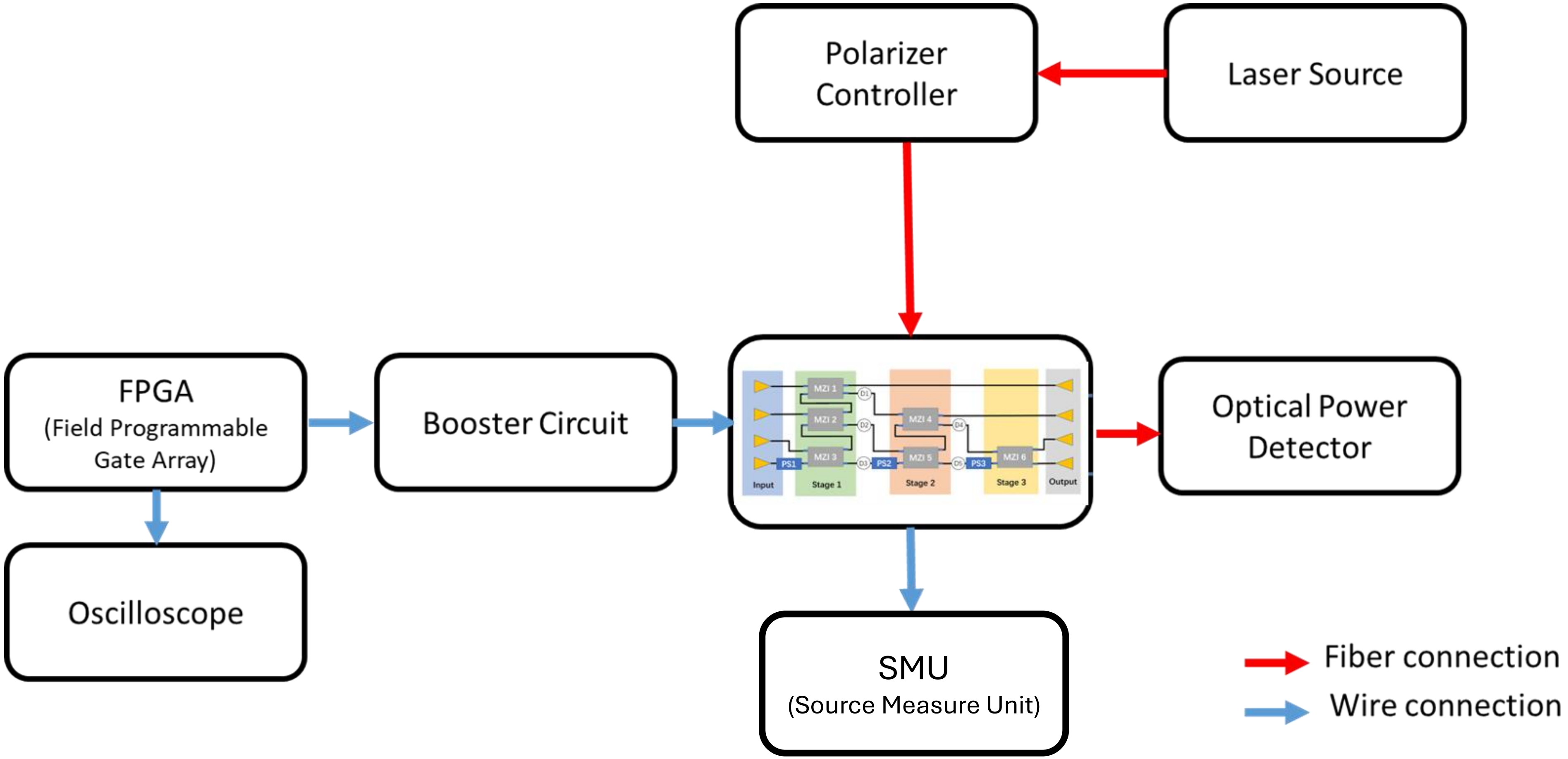
Setup


Polarizer Controller
Laser Source
FPGA
(Field Programmable Gate Array)
Booster Circuit
Optical Power Detector
Oscilloscope
SMU
(Sense Measure Unit)
Fiber connection
Wire connection
$E_{in,1}$
Port1
$E_{in,2}$
Port2
Direction Coupler
PS1
V1
PS2
V2
Direction Coupler
Port3
$E_{out,1}$
Port4
$E_{out,2}$

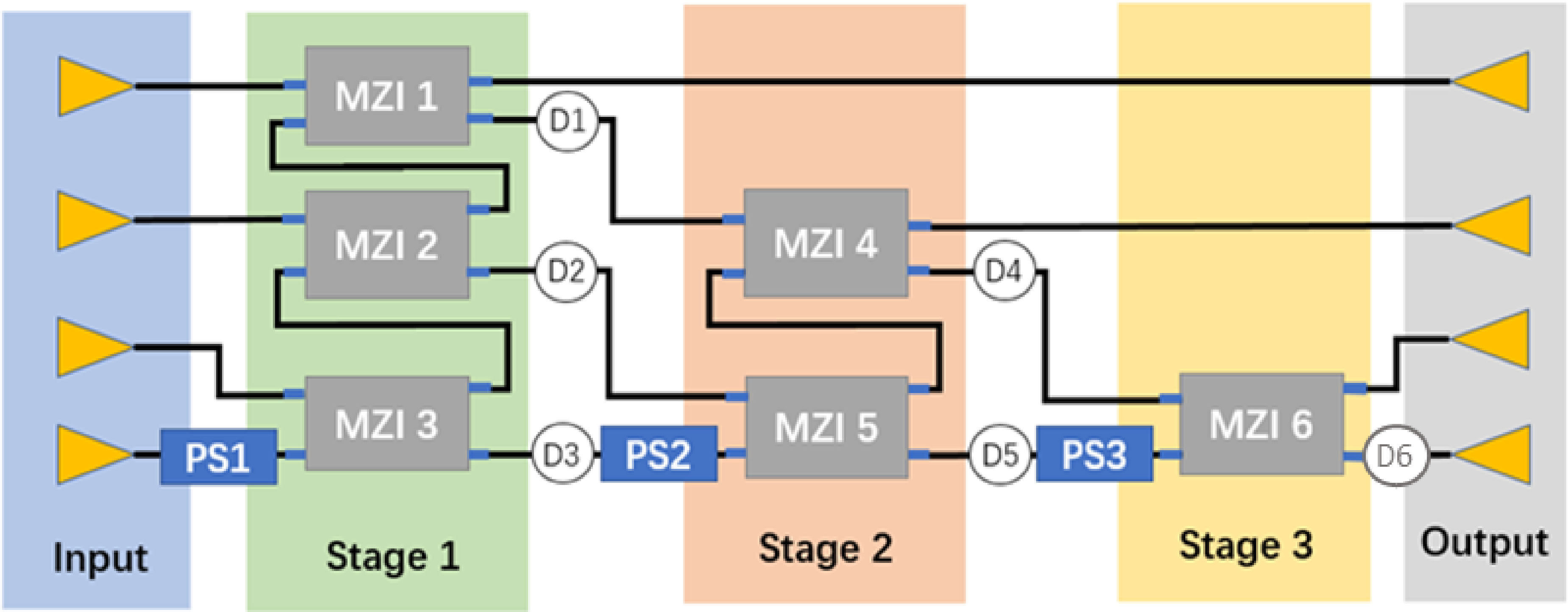

MZI1 Output bar Optical power (dBm)
DC Voltage at PS2
X 5.2635
Y -6.16073
X 4.6374
Y -9.16092
-3 dBm
-4
-6
-8
-10
-12
-14
-16
0
1
2
3
4
5
6
7
8


Output Optical power (dBm)
Consumed electrical power (W)
X 0.0547
Y -6.13253
X 0.0324
Y -13.5103
-6
-7
-8
-9
-10
-11
-12
-13
0.03
0.04
0.05
0.06
0.07
0.08
0.09

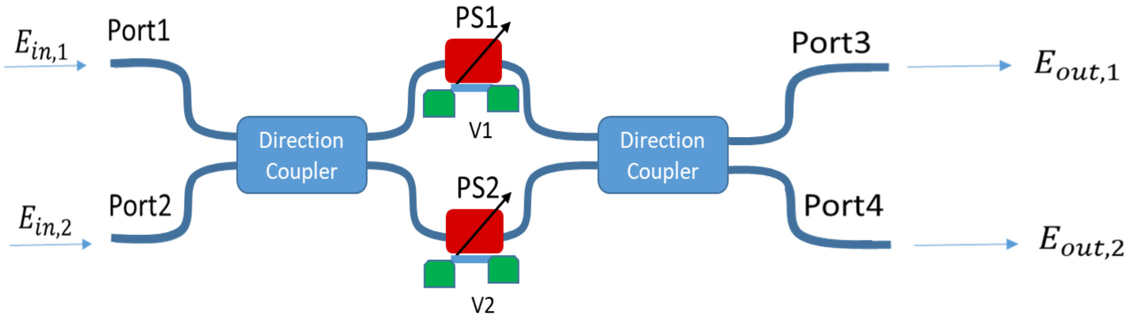
Digital Driving the Optical Phase Shift Heater

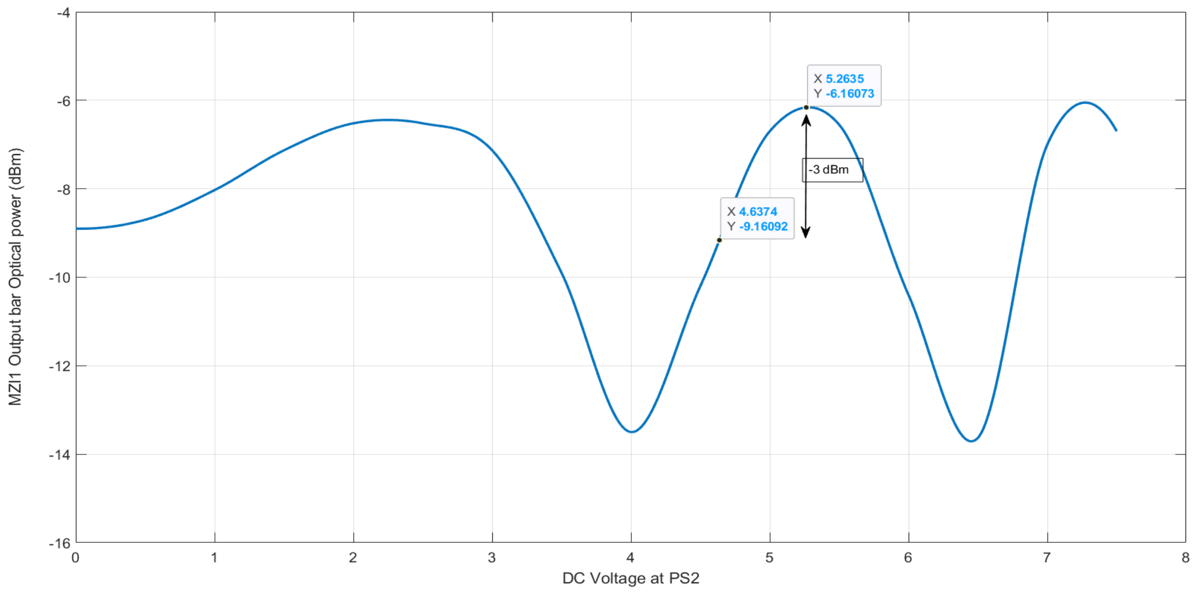


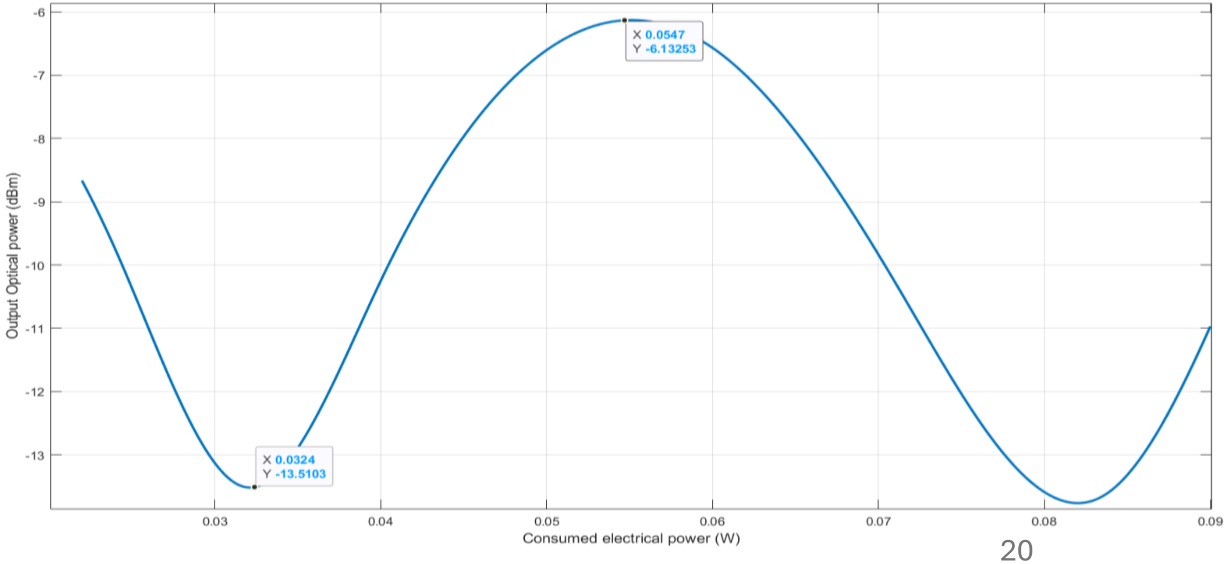
4x4 Optical Phase Shifter Circuit



Digital Driving for the Optical Phase Shift Heater





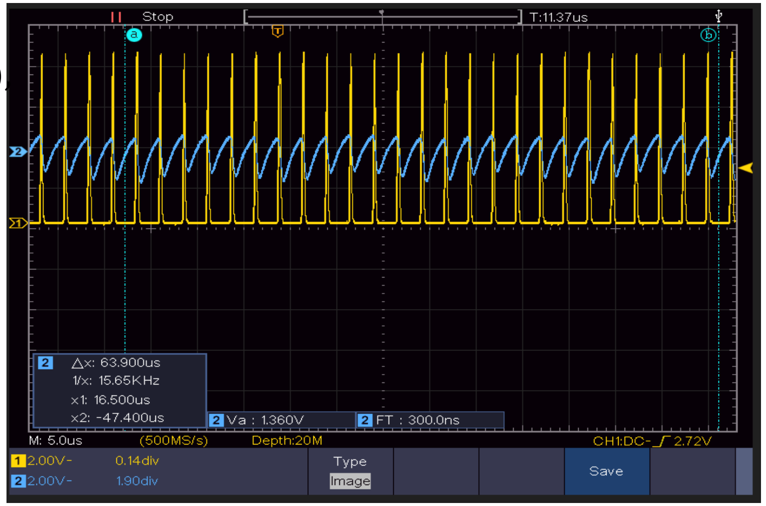


- MZI is fabricated in Cross State.
- MZI Operating voltage point @ -3dB: 4.6V
- $P_{\pi} = 34.4\ mW$

Digital Driving (FPGA) for Optical Phase Shift Heater

[Yellow]: PWM signal
f = 392 KHz (8 bits resolution),
Duty Cycle: 5% .

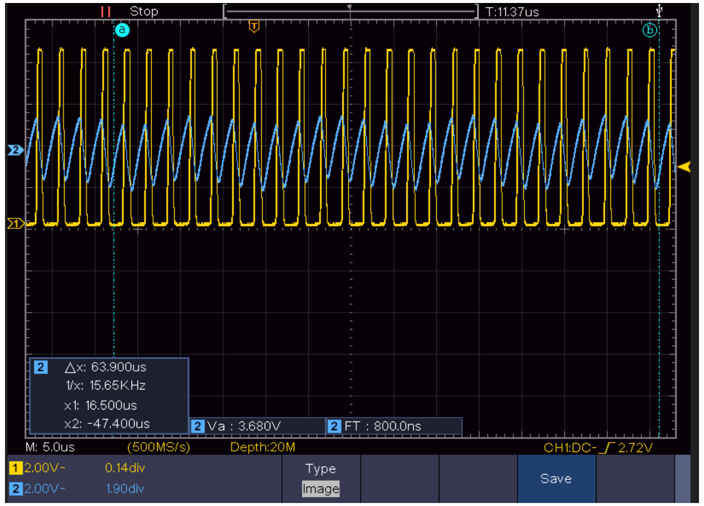
[Blue]: Photodetector Analog Output (relative phase)



[Yellow]: PWM signal

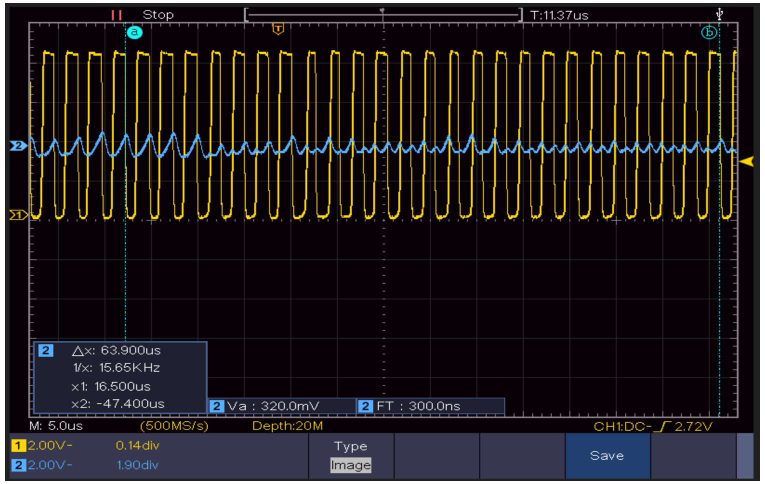
f = 392 KHz (8 bits resolution),
Duty Cycle: 10% .

[Blue]: Photodetector Analog Output (relative phase)



[Yellow]: PWM signal
f = 392 KHz (8 bits resolution),
Duty Cycle: 50% .

[Blue]: Photodetector Analog Output (relative phase)



- Phase variation issue.
  Solution? Higher frequency needed for no phase difference (steady).

- To test the MZI's heater limitation.
  Action? Function Generator was used instead of FPGA

Digital Driving (Function Generator) for Optical Phase Shift Heater

- f_{AC} = 1 KHz -> 500 KHz

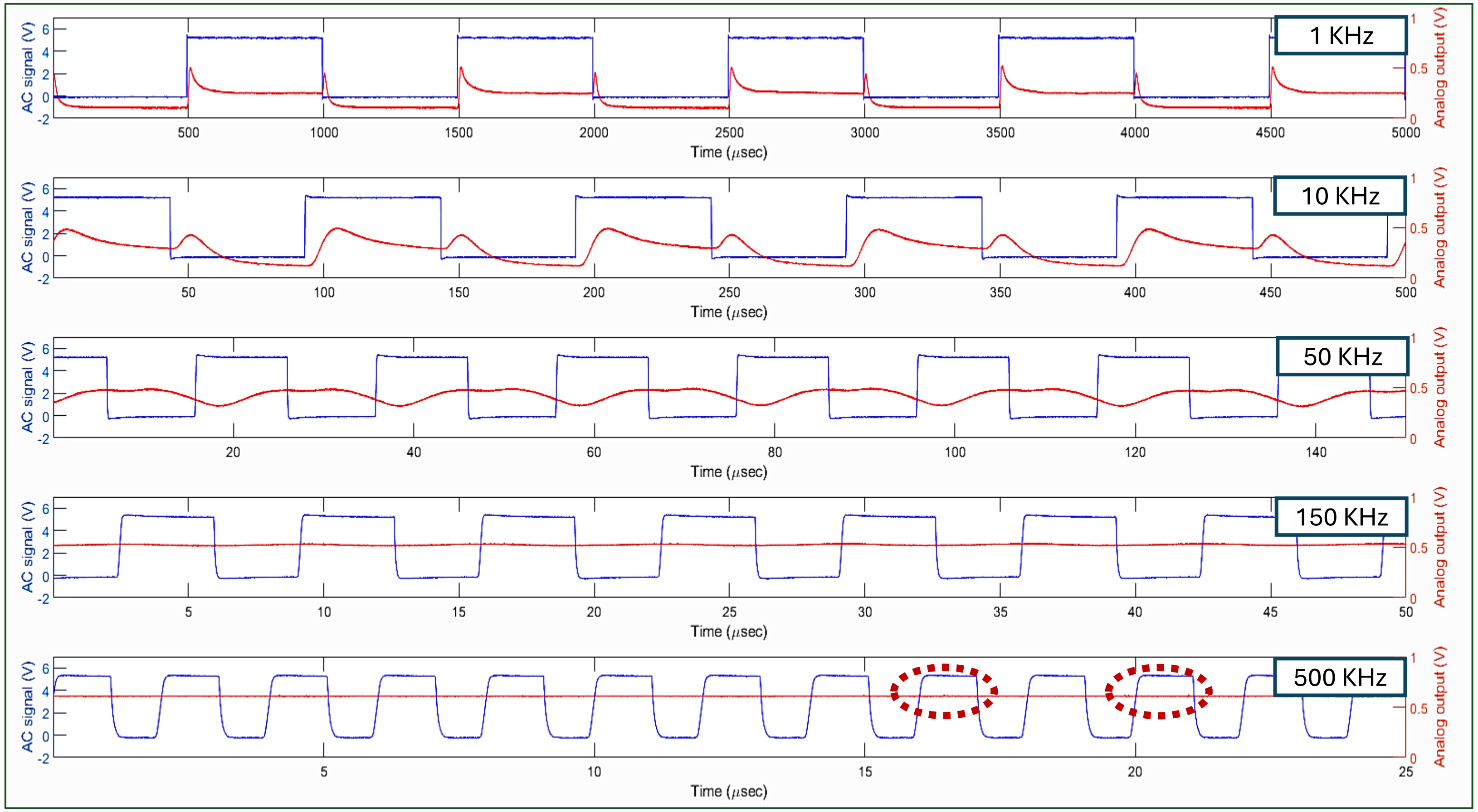
- *50% duty cycle max phase variation.*

- *Residual modulation at high frequencies, sampling precision issue.*

"Newport" meter Analog filter: 250 KHz (Datasheet)

<u>Solution?</u>

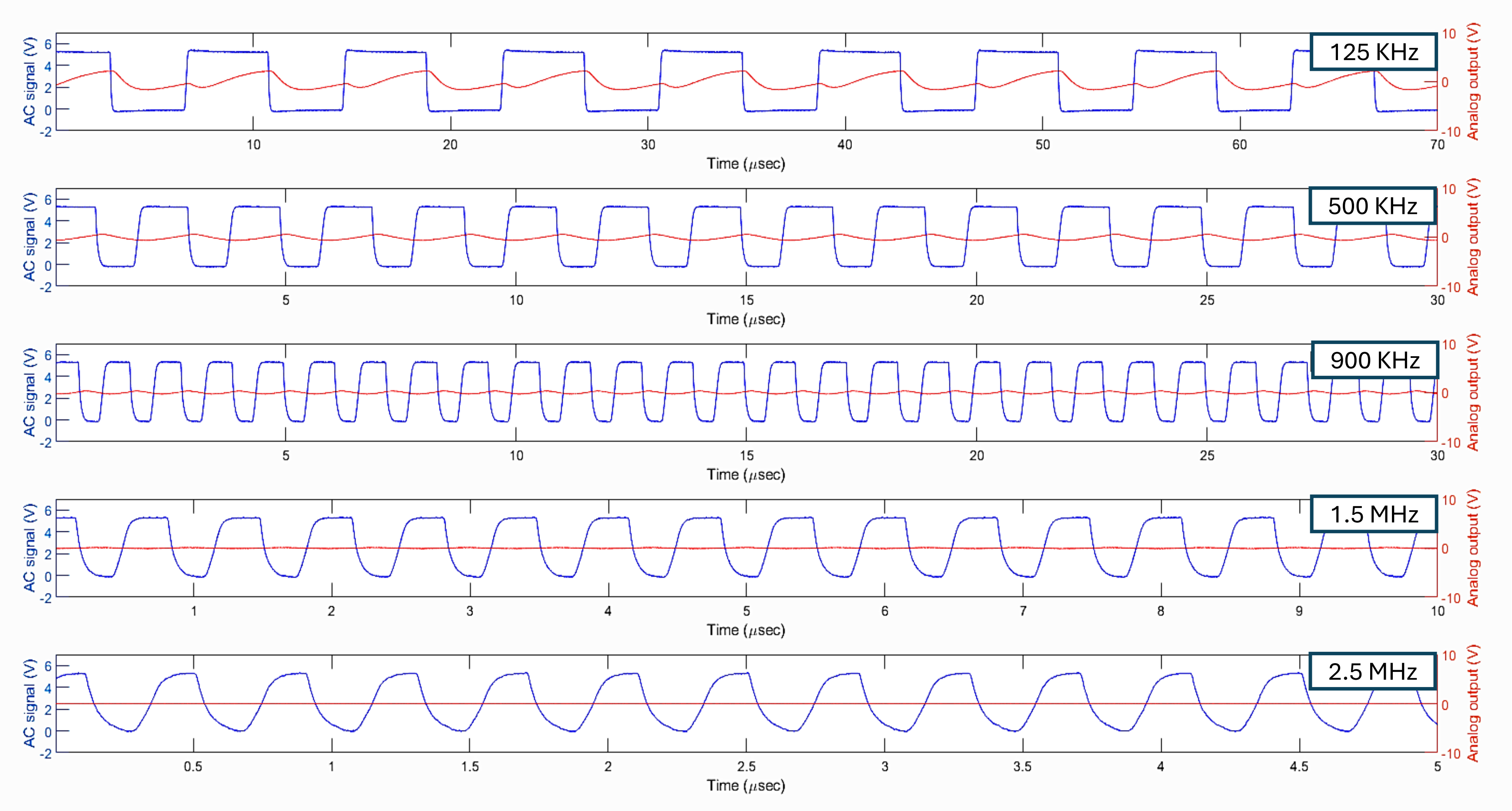
RF High-speed photodetector instead of "Newport" photodetector power meter.



Digital driving (using Function Generator) for optical phase shift heater

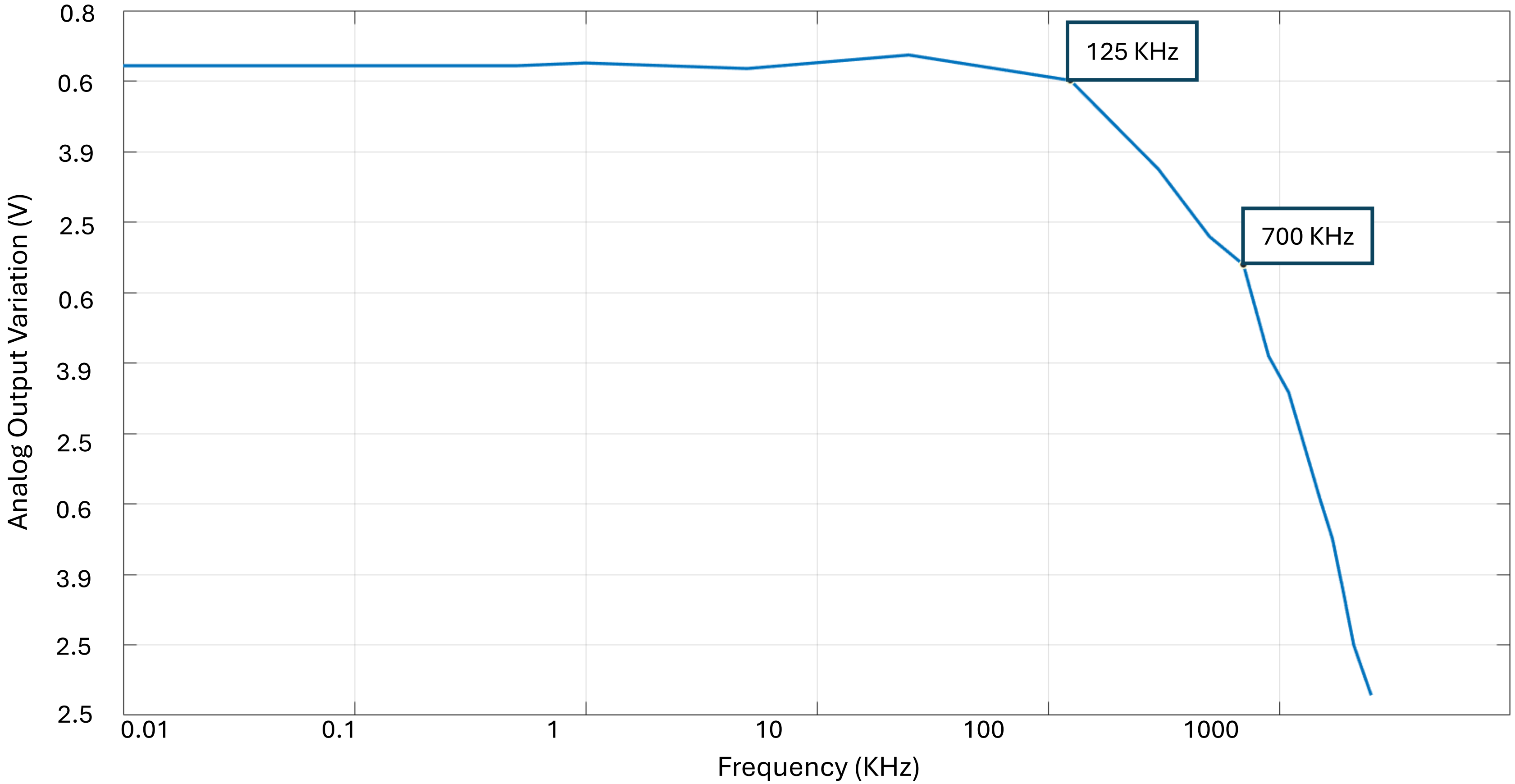
- f_{AC} = 125 KHz -> 2.5 MHz.

- *Phase with no variation = 1.5 MHz.*



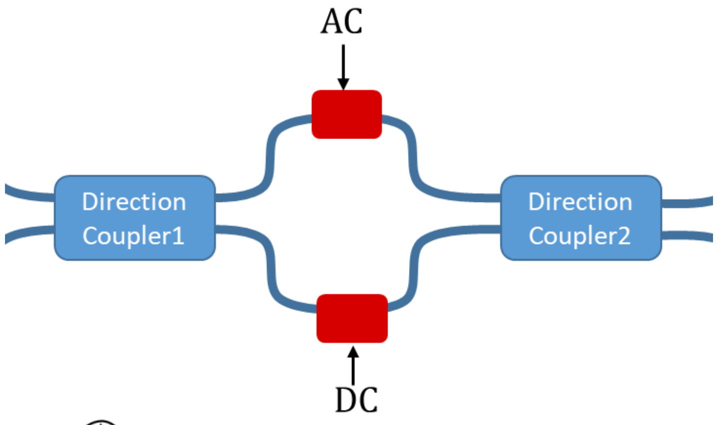
Digital driving for optical phase shift heater

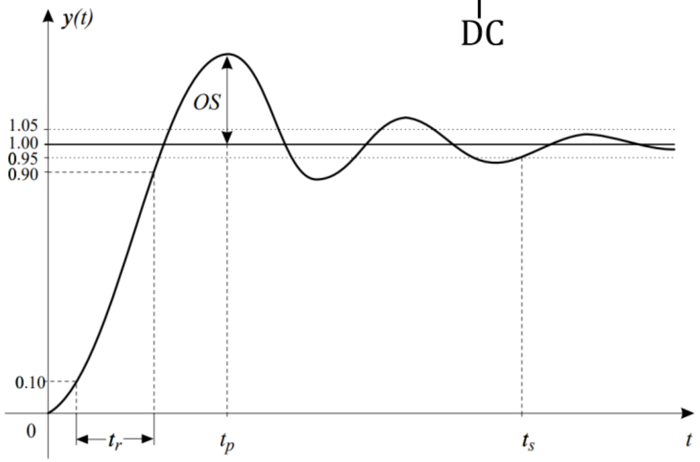
- MZI f_{AC} response, log scale.
- $f_{AC,cuttoff1} = 125\ KHz$
- $f_{AC,cuttoff2} = 700\ KHz$



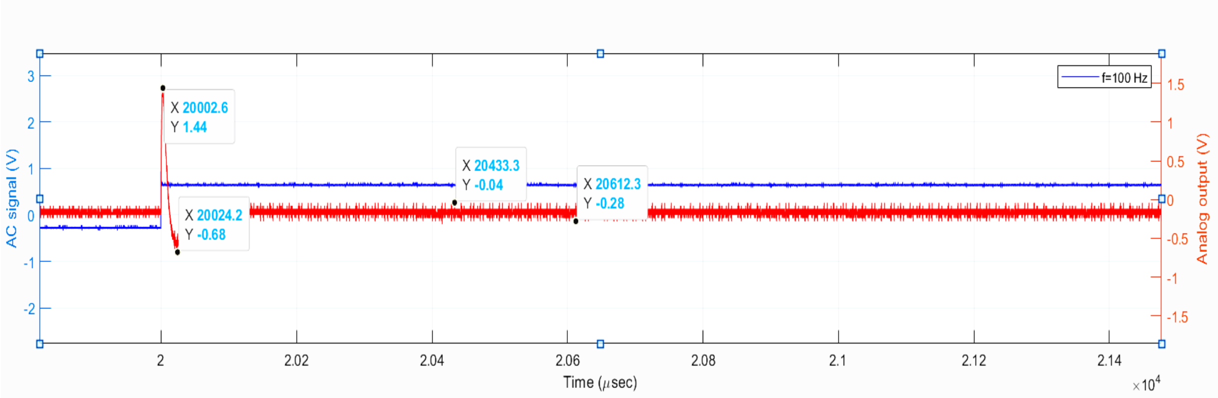
Digital driving for optical Phase Shift Heater

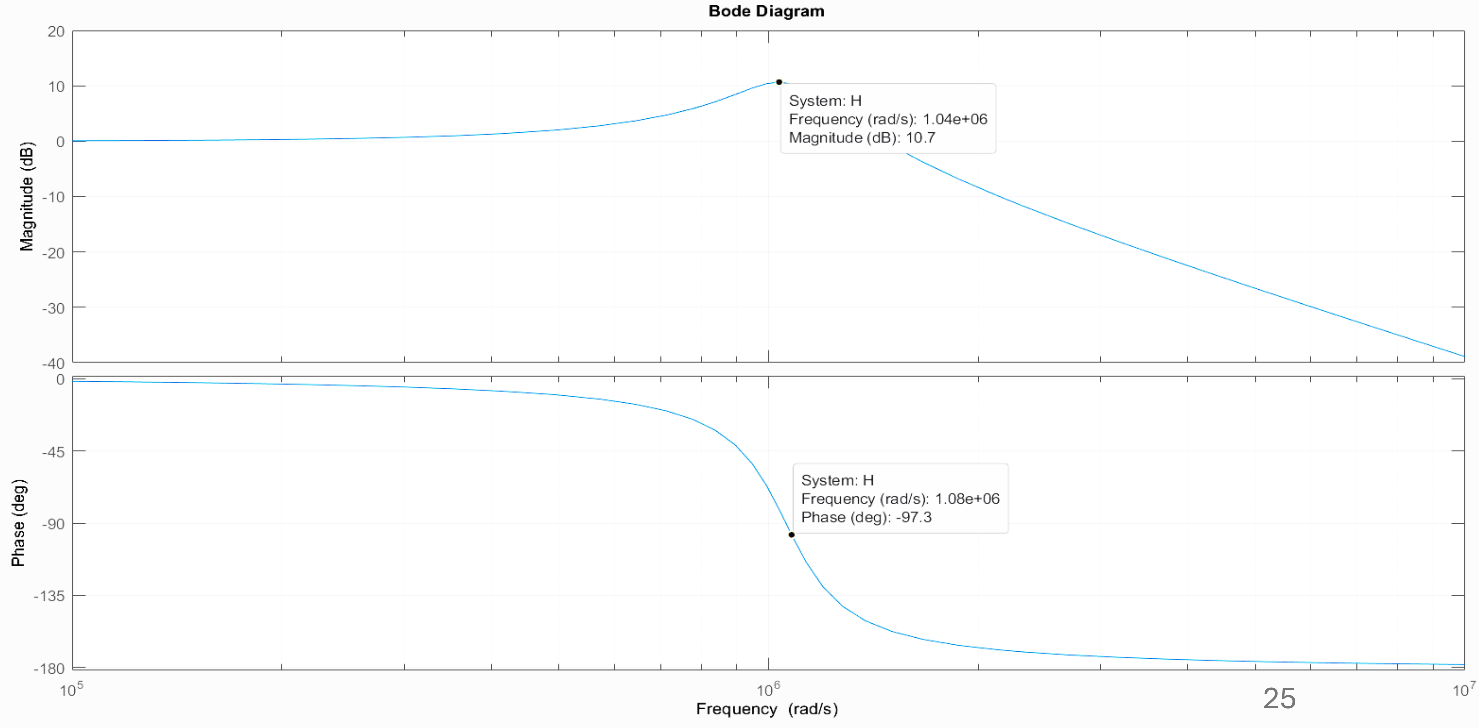
- MZI second order behavior with step response.



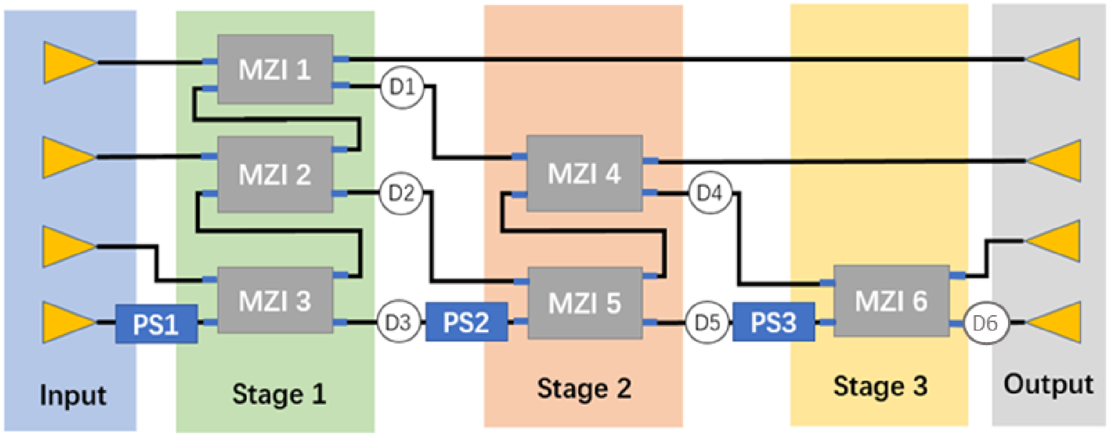


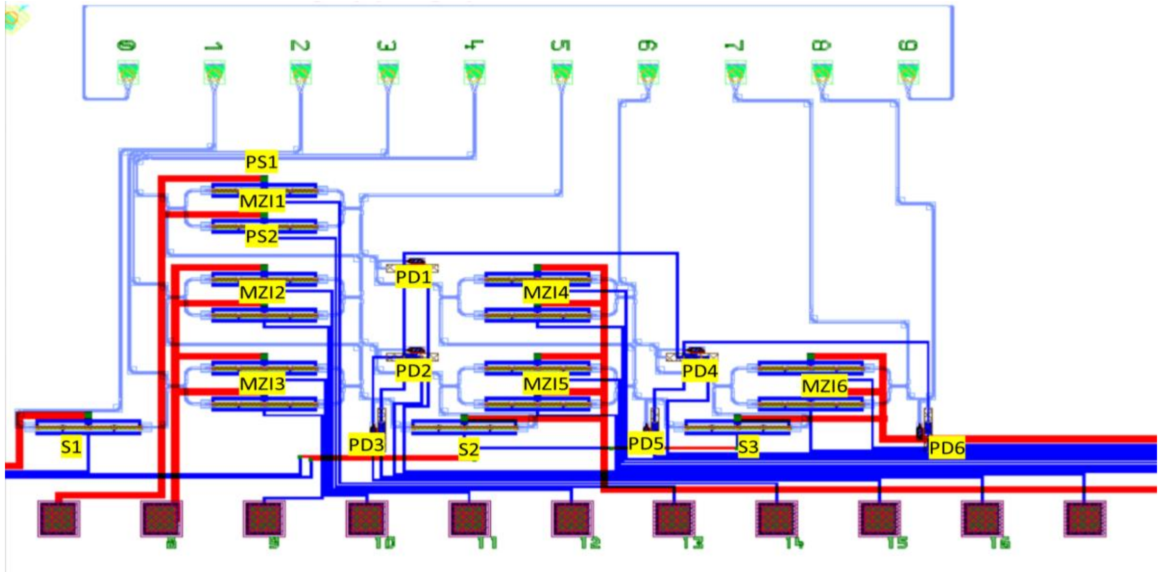
$$H(S) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{1.121e12}{S^2 + 3.133e05\ S + 1.121e12}$$





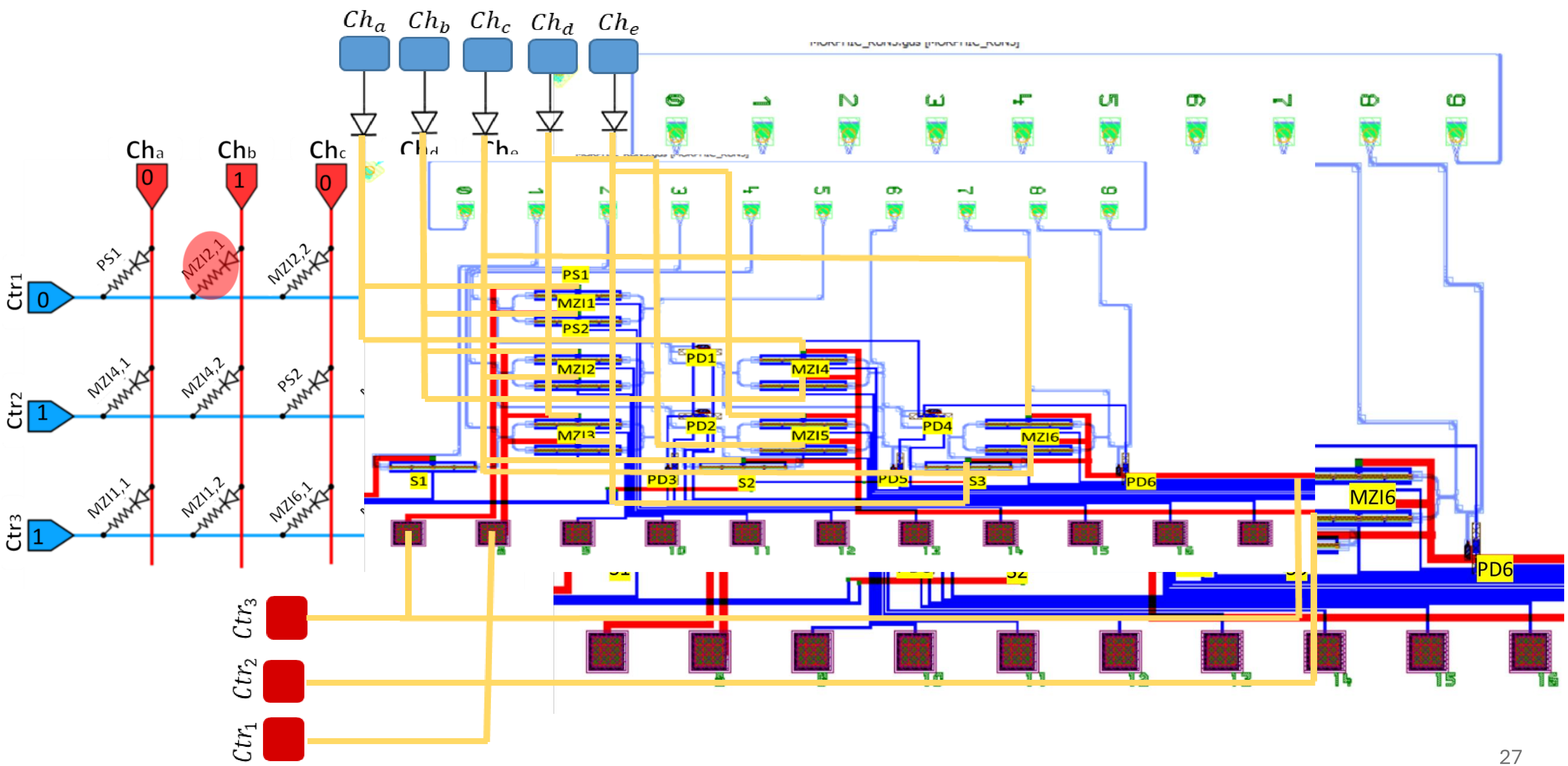
Matrix addressing configuration for the 4x4 Optical Phase Shifter Circuit





Designed by Umar Khan

Matrix addressing configuration for the 4x4 circuit



Conclusion

My contributions

- PWM digital driving using *FPGA.*
- Linear phase variation with PWM signal.
- MZI characterization (Operating voltage point and P_{π}).
- Booster circuit design.
- MZI f_{AC} step response.
- Matrix addressing circuit design.

Future Work

- Advanced FPGA board for higher PWM frequency with more resolution.
- Thermal excitation model for RC circuit response of the phase shifter heater.
- Further study for the MZI step response.
- Control the matrix addressing with FPGA.

Thank You!

Any Questions?